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Task 4: - Theory + Practical

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Sr. No.	Content
1	Cover page
2	Introduction to Network Security
3	Lab Setup
4	Network Architecture
5	Apache Installation
6	Nmap Basic Scan
7	Advanced Nmap Scan
8	Wireshark Analysis
9	Learning Outcomes
10	Conclusion



Objective:

Understand basic network setup and perform initial security testing in a lab environment.

Tools Used:

- VMware Workstation
- Kali Linux
- Ubuntu Server
- Apache Web Server
- Nmap
- Wireshark

Introduction to Network Security: -

Network security is the practice of protecting a computer network, its devices, systems, and data from unauthorized access, attacks, misuse, and damage. It includes security technologies, policies, procedures, and practices that help keep information **confidential, accurate, and available** to authorized users. Network security is important for homes, schools, businesses, banks, hospitals, government organizations, and other institutions that use computer networks.

Why Do We Need Network Security?

Network security is needed because computer networks face many types of threats, such as:

- **Unauthorized access:** Prevents attackers or unauthorized users from accessing systems and information.
- **Malware and viruses:** Helps protect computers and networks from malicious software.
- **Data theft:** Protects sensitive information such as passwords, financial information, and personal data.
- **Cyberattacks:** Helps defend against attacks such as phishing, ransomware, denial-of-service (DoS), and network intrusion.
- **Data protection:** Prevents unauthorized modification or destruction of important data.
- **Privacy:** Ensures that confidential communication and information are accessible only to authorized people.
- **Business continuity:** Helps organizations continue their operations even when security incidents occur.

Advantages of Network Security

- Protects confidential and sensitive information.
- Prevents unauthorized access to network resources.
- Reduces the risk of malware and cyberattacks.
- Improves privacy and data protection.
- Helps organizations maintain reliable network services.
- Protects devices such as computers, servers, routers, and mobile devices.
- Helps organizations comply with security policies and regulations.

Benefits of Network Security

The major benefits include:

1. **Confidentiality** – Only authorized users can access sensitive information.
2. **Integrity** – Protects data from unauthorized changes or corruption.
3. **Availability** – Keeps network services and resources accessible when needed.
4. **Authentication** – Verifies the identity of users and devices.
5. **Access control** – Determines which users can access particular resources.
6. **Risk reduction** – Reduces the possibility and impact of security incidents.
7. **Trust and reliability** – Create a safer environment for communication and sharing information.

Common Network Security Tools

Some commonly used network security tools are:

- **Firewall:** Monitors and controls incoming and outgoing network traffic based on security rules.
- **Antivirus/Anti-malware:** Detects and removes malicious software.
- **IDS (Intrusion Detection System):** Monitors network activity and alerts administrators about suspicious activity.
- **IPS (Intrusion Prevention System):** Detects and can automatically block certain malicious network activity.
- **VPN (Virtual Private Network):** Encrypts network communication and provides a secure connection over an untrusted network.
- **Wireshark:** A network protocol analyzer used to capture and examine network traffic.
- **Nmap:** A network scanning tool used to discover hosts, services, and open ports.
- **SIEM (Security Information and Event Management):** Collects and analyzes security logs and events to help identify threats.
- **Password managers:** Help users create and securely store strong passwords.

Lab Setup Explanation

Lab Architecture:-



Virtual Machines: -

Machine	Role	Example IP
Kali Linux	Attacker	192.168.188.128
Ubuntu Server	Target	192.168.188.131

Both virtual machines were connected using the same VMware NAT (or Host-Only) network.

Network Verification

The following commands were used:

Kali **Ubuntu**
`ip a` `hostname -I`

Connectivity was verified using:

- `ping 192.168.188.131`
- `ping 192.168.188.128`

The screenshot displays two terminal windows side-by-side. The left window is a Kali Linux terminal showing the output of the `ip a` command, which lists network interfaces `lo` and `eth0` with their respective IP addresses and configurations. Below this, the output of the `ping 192.168.147.130` command is shown, indicating successful connectivity with 8 packets received and 0% packet loss. The right window is an Ubuntu terminal showing the output of the `hostname -I` command, which returns `192.168.147.130`. Below this, the output of the `ping 192.168.147.129` command is shown, indicating successful connectivity with 7 packets received and 0% packet loss.

Service Installation and Configuration

Apache Web Server Installation: -

The Apache web server was installed on Ubuntu Server.

Commands Used

1. `sudo apt update`
2. `sudo apt install apache2 -y`

The service was started using:

3. `sudo systemctl start apache2`
4. `sudo systemctl enable apache2`

Service status was checked using:

5. `sudo systemctl status apache2`

The web server was tested from Kali by opening:

`http://localhost`

```
kali@kali:~$ sudo apt update
[sudo] password for kali:
Get:1 https://repo.protonvpn.com/debian stable InRelease [2,967 B]
Get:2 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:4 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.1 MB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Packages [116 kB]
Get:6 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [270 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:8 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]
Fetched 77.3 MB in 15s (5,162 kB/s)
125 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)~$ sudo apt install apache2
apache2 is already the newest version (2.4.68-2).
apache2 set to manually installed.
The following packages were automatically installed and are no longer required:
bluez-hcidump libllvmspirvlib18.1 libsodium23 openjdk-21-jre-headless
enchant-2 libmbedtlscrypto16 libteamdctl0 openssh-client-gssep1
firmware-marvell-prestera libmupdf27-0 libunibreak6 perl-modules-5.40
libbssl20260107 libntfs-3g89t64 libvidstab1.1 python3-mitmproxy-wireguard
libavc1394-0 libperl5.40 libxmlsec1-1 python3-pluginbase
libbluray3 libpoppler147 libxmlsec1-openssl1 python3-proton-vpn-local-agent
libconfig+11 libpoppler150 linux-base-6.18.12+kali-amd64 python3-t2
libconfig11 libpostl1 linux-base-6.19.14+kali-amd64 python3.13
libgdal38 libpython3.13 liblinux-binary-6.19.14+kali-amd64 python3.13-dev
libgeos3.14.1 libpython3.13-dev linux-image-6.18.12+kali-amd64 python3.13-minimal
libhdf5-hl-310 libpython3.13-minimal linux-image-6.19.14+kali-amd64 python3.13-rr
libjlm0.83 libpython3.13-stdlib linux-modules-6.19.14+kali-amd64 python3.13-venv
libkmlbase1t64 libraw23t64 llvm-spirv-18 rpcsvc-proto
libkmlcore1t64 libstdl2-classic network-manager-gnome seclists
```

```
(kali@kali)~$ apache2 -v
Server version: Apache/2.4.68 (Debian)
Server built: 2026-08-31T10:45:34
```

```
(kali@kali)~$ sudo systemctl start apache2
```

```
(kali@kali)~$ sudo systemctl enable apache2
```

```
(kali@kali)~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-09-26 03:22:49 EDT; 4min 55s ago
 Invocation: 45e2108be75c4611965fa2cde6509878
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 906 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 6 (limit: 3228)
   Memory: 36.1M (peak: 37.5M)
      CPU: 604ms
   CGroup: /system.slice/apache2.service
           └─906 /usr/sbin/apache2 -k start -DFOREGROUND
             └─993 /usr/sbin/apache2 -k start -DFOREGROUND
               └─994 /usr/sbin/apache2 -k start -DFOREGROUND
                 └─995 /usr/sbin/apache2 -k start -DFOREGROUND
                   └─996 /usr/sbin/apache2 -k start -DFOREGROUND
                     └─997 /usr/sbin/apache2 -k start -DFOREGROUND
```



Nmap Scan Analysis

Basic Network Scan

Nmap was used from Kali Linux to scan the Ubuntu Server.

Command:

```
nmap 192.168.188.131
```

The scan identified open ports.

Example Results

Port	State	Service
21	Open	ftp
80	Open	HTTP

Meaning

- Port 22 allows secure remote login.
- Port 80 hosts the Apache web server.

Advanced Nmap Analysis

Service Detection

Command:

```
nmap -sV 192.168.188.128
```

This identified running services.

Port	Service
21	ftp
53	Domain
80	Apache HTTP Server

OS Detection

Command:

```
sudo nmap -O 192.168.188.128
```

The scan detected the target operating system as Linux.

Saving Scan Results

Command:

```
nmap -sV 192.168.188.128 -oA week4_scan
```

Generated files:

- week4_scan.nmap
- week4_scan.xml
- week4_scan.gnmap

```
(kali@kali) [~/Desktop/week4]
$ nmap -sV -p 192.168.188.131/24 -oA full_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 04:30 -0400
Stats: 0:00:03 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 90.39% done; ETC: 04:30 (0:00:00 remaining)
Stats: 0:00:04 elapsed; 250 hosts completed (4 up), 4 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.17% done
Stats: 0:00:04 elapsed; 250 hosts completed (4 up), 4 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 0.60% done
Stats: 0:00:04 elapsed; 250 hosts completed (4 up), 4 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 1.37% done; ETC: 04:32 (0:01:12 remaining)
Nmap scan report for 192.168.188.1
Host is up (0.00076s latency).
All 65535 scanned ports on 192.168.188.1 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)
MAC Address: 00:50:56:C0:00:08 (VMware)

Nmap scan report for 192.168.188.2
Host is up (0.00044s latency).
Not shown: 65534 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
53/tcp    open  dnsmasq 2.45
MAC Address: 00:50:56:FD:96:40 (VMware)

Nmap scan report for 192.168.188.131
Host is up (0.00059s latency).
All 65535 scanned ports on 192.168.188.131 are in ignored states.
Not shown: 65535 closed tcp ports (reset)
MAC Address: 00:0C:29:A3:15:47 (VMware)

Nmap scan report for 192.168.188.254
Host is up (0.00024s latency).
All 65535 scanned ports on 192.168.188.254 are in ignored states.
Not shown: 65535 filtered tcp ports (no-response)
MAC Address: 00:50:56:EA:89:F5 (VMware)

Nmap scan report for 192.168.188.128
Host is up (0.00021s latency).
Not shown: 65533 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
80/tcp    open  http     Apache httpd 2.4.68 ((Debian))
Service Info: OS: Unix

(kali@kali) [~/Desktop/week4]
$ sudo nmap -O 192.168.188.131/24 -oN OS_detection.txt
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-26 04:33 -0400
Nmap scan report for 192.168.188.1
Host is up (0.0035s latency).
All 1000 scanned ports on 192.168.188.1 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:50:56:C0:00:08 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

Nmap scan report for 192.168.188.2
Host is up (0.0017s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 00:50:56:FD:96:40 (VMware)
Device type: specialized|general purpose
Running (JUST GUESSING): VMware Player (90%), Microsoft Windows XP/7/2012 (86%)
OS CPE: cpe:/a:vmware:player cpe:/o:microsoft:windows_xp::sp3 cpe:/o:microsoft:windows_xp::sp3
Aggressive OS guesses: VMware Player virtual NAT device (90%), Microsoft Windows XP/7/2012 (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

Nmap scan report for 192.168.188.131
Host is up (0.00099s latency).
All 1000 scanned ports on 192.168.188.131 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:A3:15:47 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop
```

Nmap Findings

Open Ports

Port	Service	Risk
21	ftp	Medium
53	Domain	Medium
80	HTTP	Medium

Observations

- DNS service was available on port 53.
- FTP service was available on port 21.
- Apache HTTP service was available on port 80.
- Nmap successfully detected the running services and open ports.

Wireshark Traffic Analysis

Packet Capture

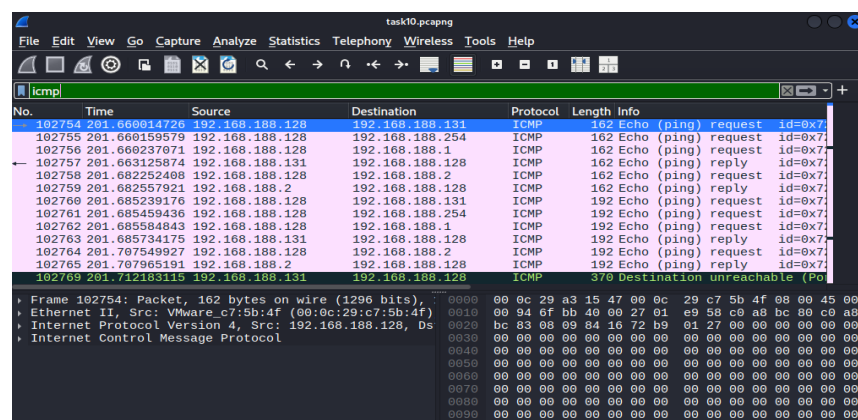
Wireshark was used on Kali Linux to capture network traffic.

During capture:

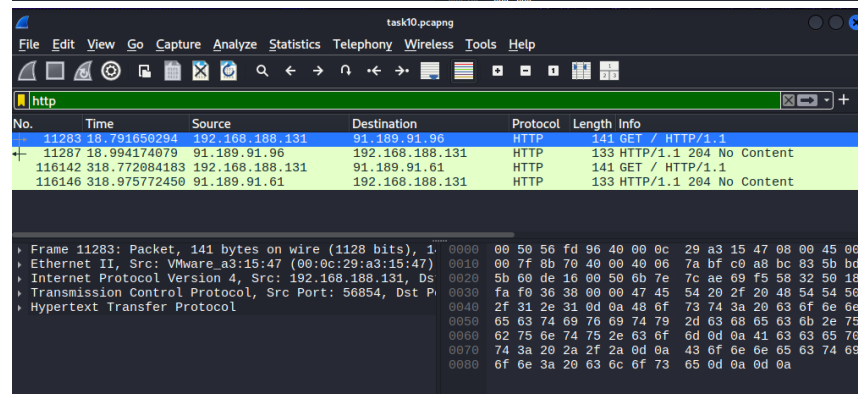
- Apache webpage was opened.
- Ping was executed.
- DNS requests were generated.

The captured traffic included:

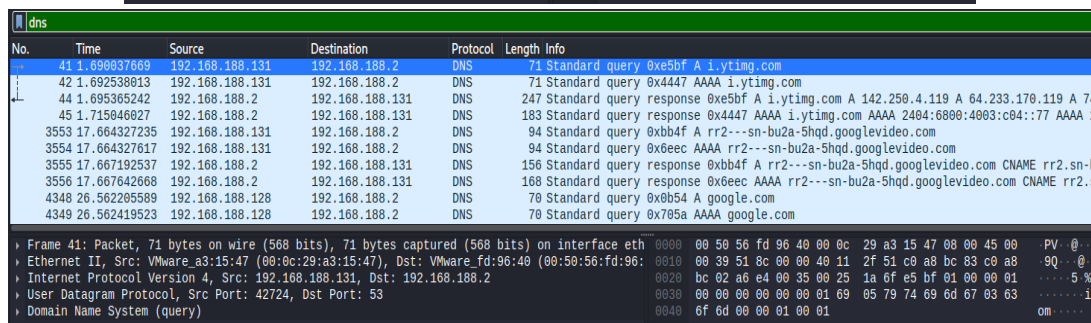
- HTTP
- DNS
- ICMP



No.	Time	Source	Destination	Protocol	Length	Info
102754	201.660014720	192.168.188.128	192.168.188.131	ICMP	162	Echo (ping) request id=0x7f
102755	201.660159579	192.168.188.128	192.168.188.254	ICMP	162	Echo (ping) request id=0x7f
102756	201.660237071	192.168.188.128	192.168.188.1	ICMP	162	Echo (ping) request id=0x7f
102757	201.660325874	192.168.188.131	192.168.188.128	ICMP	162	Echo (ping) reply id=0x7f
102758	201.682252408	192.168.188.128	192.168.188.2	ICMP	162	Echo (ping) request id=0x7f
102759	201.682557921	192.168.188.2	192.168.188.128	ICMP	162	Echo (ping) reply id=0x7f
102760	201.685239176	192.168.188.128	192.168.188.131	ICMP	192	Echo (ping) request id=0x7f
102761	201.685459436	192.168.188.128	192.168.188.254	ICMP	192	Echo (ping) request id=0x7f
102762	201.685584843	192.168.188.128	192.168.188.1	ICMP	192	Echo (ping) request id=0x7f
102763	201.685734175	192.168.188.131	192.168.188.128	ICMP	192	Echo (ping) reply id=0x7f
102764	201.707549927	192.168.188.128	192.168.188.2	ICMP	192	Echo (ping) request id=0x7f
102765	201.707965191	192.168.188.2	192.168.188.128	ICMP	192	Echo (ping) reply id=0x7f
102769	201.712183115	192.168.188.131	192.168.188.128	ICMP	370	Destination unreachable (Po



No.	Time	Source	Destination	Protocol	Length	Info
11283	18.791656294	192.168.188.131	91.189.91.96	HTTP	141	GET / HTTP/1.1
11287	19.904174079	91.189.91.96	192.168.188.131	HTTP	133	HTTP/1.1 204 No Content
116142	318.772084183	192.168.188.131	91.189.91.61	HTTP	141	GET / HTTP/1.1
116146	318.975772450	91.189.91.61	192.168.188.131	HTTP	133	HTTP/1.1 204 No Content



No.	Time	Source	Destination	Protocol	Length	Info
41	1.690037669	192.168.188.131	192.168.188.2	DNS	71	Standard query 0xe5bf A i.ytimg.com
42	1.692538013	192.168.188.131	192.168.188.2	DNS	71	Standard query 0x4447 AAAA i.ytimg.com
44	1.695365242	192.168.188.2	192.168.188.131	DNS	247	Standard query response 0xe5bf A i.ytimg.com A 142.250.4.119 A 64.233.170.119 A 7
45	1.715046027	192.168.188.2	192.168.188.131	DNS	183	Standard query response 0x4447 AAAA i.ytimg.com AAAA 2404:6800:4003:c04::77 AAAA
3553	17.664327235	192.168.188.131	192.168.188.2	DNS	94	Standard query 0xbb4f A rr2---sn-bu2a-5hqdg.googlevideo.com
3554	17.664327617	192.168.188.131	192.168.188.2	DNS	94	Standard query 0x6eec AAAA rr2---sn-bu2a-5hqdg.googlevideo.com
3555	17.667192537	192.168.188.2	192.168.188.131	DNS	156	Standard query response 0xbb4f A rr2---sn-bu2a-5hqdg.googlevideo.com CNAME rr2-sn-b
3556	17.667642668	192.168.188.2	192.168.188.131	DNS	168	Standard query response 0x6eec AAAA rr2---sn-bu2a-5hqdg.googlevideo.com CNAME rr2-sn-b
4348	26.562205589	192.168.188.128	192.168.188.2	DNS	70	Standard query 0xb054 A google.com
4349	26.562419523	192.168.188.128	192.168.188.2	DNS	70	Standard query 0x705a AAAA google.com

Learning Outcomes and References

Learning Outcomes: -

During this lab I learned:

- How to create a virtual network in VMware.
- How to connect multiple virtual machines.
- How to install and configure Apache.
- How to perform Nmap scanning.
- How to identify open ports and services.
- How to capture network traffic using Wireshark.
- How to analyze HTTP, DNS, and ICMP packets.
- How to save scan results and packet capture files.

References: -

- Nmap Official Documentation → <https://nmap.org/docs.html>
- Wireshark Official Documentation → <https://www.wireshark.org/docs/relnotes/>
- Apache HTTP Server Documentation → <https://httpd.apache.org/docs/>
- Ubuntu Documentation → <https://ubuntu.com/desktop/docs/en/latest/>
- Kali Linux Documentation → <https://www.kali.org/docs/>

Conclusion

- Successfully created a virtual network using VMware.
- Connected Kali Linux and Ubuntu Server on the same network.
- Verified network connectivity using the ping command.
- Installed and configured the Apache Web Server on Ubuntu.
- Confirmed that the web server was running properly.
- Used Nmap to identify open ports on the target machine.
- Detected running services such as Domain and Apache.
- Captured network traffic using Wireshark.
- Analyzed HTTP, DNS, and ICMP packets during communication.
- Gained practical experience in basic network security testing and traffic analysis in a safe lab environment.